
Analytical Derivation of the Geodesic Equation for the 2-Sphere

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1 Intuition

To determine shortest paths on a non-Euclidean manifold, the geodesic equation can be used, incorporating the curvature. On the 2-sphere these shortest paths lie on great circles—something that becomes intuitive when visualized.

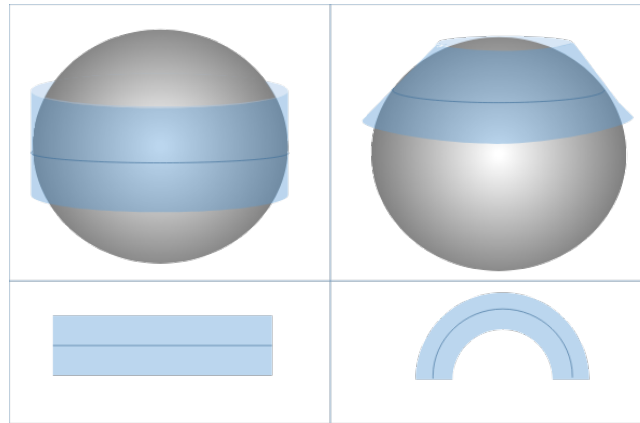


Figure 1: Great circle vs small circle

A great circle can be drawn on a sphere by wrapping a rectangle around it on which the line is drawn. Unwrapping the rectangle results in a straight line on the paper. For any circle that is not a great circle, the rectangular piece of paper must be folded or warped into a ring in order to draw it. When unwrapped, it shows a curved line. Thus, any path between two points on a sphere that is not part of a great circle is intuitively not the shortest path.

2 Expectation

The expected outcome for the solution of the geodesic equation on a sphere is a zero-centered plane that intersects the sphere. The line of intersection is exactly the great circle, which is the geodesic.

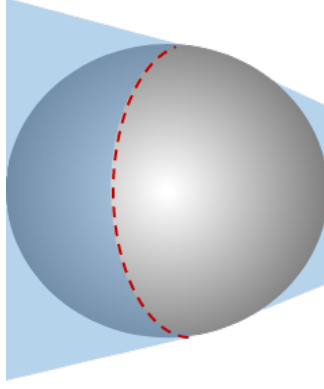


Figure 2: Intersection plane with zero centered sphere resulting in great cricle.

3 Analytical Derivation

The parameter form of the s2-sphere:

$$P(\theta, \phi) = \begin{cases} x = R \sin \theta \cos \phi, \\ y = R \sin \theta \sin \phi, \\ z = R \cos \theta. \end{cases} \quad (3.1)$$

And the partial derivative by θ and ϕ :

$$\frac{\partial P(\theta, \phi)}{\partial \theta} = \begin{cases} x = R \cos \theta \cos \phi, \\ y = R \cos \theta \sin \phi, \\ z = -R \sin \theta \end{cases} \quad (3.2) \quad \frac{\partial P(\theta, \phi)}{\partial \phi} = \begin{cases} x = -R \sin \theta \sin \phi, \\ y = R \sin \theta \cos \phi, \\ z = 0 \end{cases} \quad (3.3)$$

The distance on a Riemannian manifold is generally defined as: $ds^2 = g_{nm} dx^n dx^m$ where g_{nm} is defined as the covariant metric tensor (which is the one, we are using in the subsequent calculations, thus the attribute "covariant" will be neglected in the following) and x^n and x^m are contravariant coordinates of the sphere. Those will be replaced by the polar- and azimuth angle θ and ϕ

$$g_{nm} = \begin{pmatrix} \frac{\partial P(\theta, \phi)}{\partial \theta} \cdot \frac{\partial P(\theta, \phi)}{\partial \theta} & \frac{\partial P(\theta, \phi)}{\partial \theta} \cdot \frac{\partial P(\theta, \phi)}{\partial \phi} \\ \frac{\partial P(\theta, \phi)}{\partial \phi} \cdot \frac{\partial P(\theta, \phi)}{\partial \theta} & \frac{\partial P(\theta, \phi)}{\partial \phi} \cdot \frac{\partial P(\theta, \phi)}{\partial \phi} \end{pmatrix} \quad (3.4)$$

Which results in:

$$g_{nm} = \begin{pmatrix} R^2 & 0 \\ 0 & R^2 \sin^2 \theta \end{pmatrix} \quad (3.5)$$

Thus, d^2 can be written as:

$$ds^2 = R^2(d\theta^2 + \sin^2 \theta d\phi^2) \quad (3.6)$$

To calculate the geodesic equation, the variational principle can be used, since a geodesic can be considered the shortest path between two points on a manifold. The curvature of the manifold is implicitly taken into account through the metric tensor, and thus the Christoffel symbols do not need to be computed explicitly. To apply the variational principle to the geodesic equation, a Lagrangian must be defined in order to use the Euler–Lagrange equations.

The goal of the Euler-Lagrange equation is to minimize a functional of the form:

$$I[q] = \int F[t, q(x), q'(x)] dt \quad (3.7)$$

Which looks the following for the geodesic equation on the sphere, if θ is parametrized by ϕ :

$$I[\theta] = \int F[\phi, \theta(\phi), \theta'(\phi)] d\phi \quad (3.8)$$

The general form of the Euler Lagrange equation looks the following:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0 \quad (3.9)$$

The parameterization variable is usually named t but is not necessarily connected to the time, as in equations from physics. If the chosen Lagrange function does not directly depend on the parameter t , a short version can be used, called the Beltrami Identity, where C is a constant.

$$L - \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} = C \quad (3.10)$$

Which is for the application on the sphere:

$$L - \dot{\theta} \frac{\partial L}{\partial \dot{\theta}} = C \quad (3.11)$$

For solving this equation, the following steps are necessary:

$$\frac{\partial L}{\partial \dot{\theta}} = R \frac{1}{2} (\dot{\theta}^2 + \sin^2 \theta)^{-1/2} 2\dot{\theta} = \frac{R\dot{\theta}}{\sqrt{\dot{\theta}^2 + \sin^2 \theta}} \quad (3.12)$$

$$\frac{\partial L}{\partial \theta} = R \frac{1}{2} (\dot{\theta}^2 + \sin^2 \theta)^{-1/2} 2 \cos \theta \sin \theta = \frac{R \sin \theta \cos \theta}{\sqrt{\dot{\theta}^2 + \sin^2 \theta}} \quad (3.13)$$

Those results can then be put to the Euler Lagrange equation:

$$R \sqrt{\dot{\theta}^2 + \sin^2 \theta} - \frac{\dot{\theta}^2 R}{\sqrt{\dot{\theta}^2 + \sin^2 \theta}} = C \quad (3.14)$$

$$\frac{\dot{\theta}^2 + \sin^2 \theta - \dot{\theta}^2}{\sqrt{\dot{\theta}^2 + \sin^2 \theta}} = \frac{C}{R} \quad (3.15)$$

$$\frac{\sin^2 \theta}{\sqrt{\dot{\theta}^2 + \sin^2 \theta}} = \frac{C}{R} = k \quad (3.16)$$

Where k is a constant replacing $\frac{C}{R}$.

This result can be solved by $\dot{\theta}$

$$\dot{\theta}^2 = \frac{\sin^2 \theta}{k^2} (\sin^2 \theta - k^2) \quad (3.17)$$

Next, a short but effective substitution can be applied: $u = \cot \theta$ with $\frac{du}{d\phi} = -\frac{1}{\sin^2 \theta} \frac{d\theta}{d\phi}$ since the derivative of $\cot(x)$ is $-\csc^2(x)$ which is $\frac{1}{\sin^2(x)}$.

Thus, $\dot{\theta}^2$ from equation 3.17 can be rewritten as $(\frac{du}{d\phi})^2$:

$$(\frac{du}{d\phi})^2 = \frac{1}{\sin^4 \theta} \frac{\sin^2 \theta}{k^2} (\sin^2 \theta - k^2) = \frac{1}{k^2} (1 - \frac{k^2}{\sin^2 \theta}) \quad (3.18)$$

Since $\frac{1}{\sin^2(x)} = 1 + \cot^2(x)$:

$$\left(\frac{du}{d\phi}\right)^2 = \frac{1}{k^2} (1 - k^2(1 + \cot^2 \theta)) \quad (3.19)$$

$$= \frac{1}{k^2} (1 - k^2(1 + u^2)) \quad (3.20)$$

$$= \frac{1}{k^2} (1 - k^2 + k^2 u^2) \quad (3.21)$$

$$= \frac{1}{k^2} - 1 - u^2 \quad (3.22)$$

$$= \frac{1 - k^2}{k^2} - u^2 \quad (3.23)$$

$$= \left(\frac{\sqrt{1 - k^2}}{k}\right)^2 - u^2 \quad (3.24)$$

Since the goal is to reach a standard form like $A^2 - u^2$, $A^2 = \frac{1 - k^2}{k^2} \rightarrow A = \frac{\sqrt{1 - k^2}}{k}$. This summarizes the constant expression involving k by introducing A .

Thus:

$$\frac{du}{d\phi} = + - \sqrt{A^2 - u^2} \quad (3.25)$$

Now the equation can be separated with everything concerning u on the left and ϕ on the right:

$$\int \frac{du}{\sqrt{A^2 - u^2}} = + - \int d\phi \quad (3.26)$$

According to the standard form, the integral of 3.26 can be solved in the following way, using ϕ_0 as an integration constant:

$$\arcsin\left(\frac{u}{A}\right) = + - (\phi - \phi_0) \quad (3.27)$$

Resubstitution of $u = \cot\theta$

$$\arcsin\left(\frac{\cot\theta}{A}\right) = + - (\phi - \phi_0) \quad (3.28)$$

A short redefinition of ϕ_0 will remove the $+ -$. Since ϕ goes from $-\pi$ to $+\pi$, π can be added if ϕ is negative, which shifts the range from 0 to 2π

$$\phi' = \begin{cases} \phi & \text{if positive} \\ \phi + \pi & \text{if negative} \end{cases} \quad (3.29)$$

Isolating theta and replacing ϕ_0 by ϕ' leads to the following:

$$\cot\theta = A \sin(\phi - \phi') \quad (3.30)$$

This result is certainly not the expected equation of a plane through the center of the sphere. To end up with a plane of the form $ax+bx+cz$ further adjustments are necessary.

From the parameter form of the s2 sphere z can be taken as $R\cos\theta$. Furthermore, the definition of the $\cot\theta$ is known as $\frac{\cos\theta}{\sin\theta}$. If the sphere is projected to the x-y plane then, the distance to the z-axis can be calculated as $r = \sqrt{x^2 + y^2}$ which is simply the radius for a point at x,y.

$$r = \sqrt{(R \sin\theta \cos\phi)^2 + (R \sin\theta \sin\phi)^2} = R \sin\theta \quad (3.31)$$

The formula for r looks very similar to the formula for z. This leads to one final step:

$$\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{R \cos\theta}{R \sin\theta} = \frac{z}{r} \quad (3.32)$$

This can now be applied to equation 3.30.

$$\frac{z}{\sqrt{x^2 + y^2}} = A \sin(\phi - \phi') \quad (3.33)$$

$$\frac{z}{\sqrt{x^2 + y^2}} = A(\sin\phi \cos\phi' - \cos\phi \sin\phi') \quad (3.34)$$

It can also be derived from the projection to the x-y plane that $\cos\phi = \frac{x}{r}$ and $\sin\phi = \frac{y}{r}$. Replacing $\cos\phi$ and $\sin\phi$ in 3.34 with those fractions leads to:

$$\frac{z}{\sqrt{x^2 + y^2}} = \frac{A y \cos\phi'}{\sqrt{x^2 + y^2}} - \frac{A x \sin\phi'}{\sqrt{x^2 + y^2}} \quad (3.35)$$

$$A y \cos\phi' - A x \sin\phi' - z = 0 \quad (3.36)$$

$$A x \sin\phi' + A y \cos\phi' + z = 0 \quad (3.37)$$

To finalize the equation for the plane, set $A \cos \phi' = \alpha$ and $A \sin \phi' = \beta$

$$\alpha x - \beta y + z = 0 \quad (3.38)$$

Having obtained the equation of the plane cutting the sphere into halves and so creating a great circle at a certain point of the curve, the final question is: What is α and what is β ?

We can now rewrite the formula 3.30 to use α and β .

$$\cot \theta = A \sin(\phi - \phi') = A \sin \phi \cos \phi' - A \cos \phi \sin \phi' = \alpha \sin \phi - \beta \cos \phi \quad (3.39)$$

Assuming there are two non-identical points $p_1 = (\theta_1, \phi_1)$ and $p_2 = (\theta_2, \phi_2)$ two equations can be set:

$$\cot \theta_1 = \alpha \sin \phi_1 - \beta \cos \phi_1 \quad (3.40)$$

$$\cot \theta_2 = \alpha \sin \phi_2 - \beta \cos \phi_2 \quad (3.41)$$

This can be solved using Cramer's Rule:

$$A = \begin{pmatrix} \sin \phi_1 & -\cos \phi_1 \\ \sin \phi_2 & -\cos \phi_2 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} \cot \theta_1 \\ \cot \theta_2 \end{pmatrix} \quad (3.42)$$

This leads to the following determinants:

$$D = -\cos \phi_2 \sin \phi_1 + \cos \phi_1 \sin \phi_2 \quad (3.43)$$

$$D_1 = -\cos \phi_2 \cot \theta_1 + \cos \phi_1 \cot \theta_2 \quad (3.44)$$

$$D_2 = \sin \phi_1 \cot \theta_2 - \sin \phi_2 \cot \theta_1 \quad (3.45)$$

$$(3.46)$$

Thus, α and β are:

$$\alpha = \frac{D_1}{D} = \frac{\cot \theta_1 \cos \phi_2 - \cot \theta_2 \cos \phi_1}{\sin(\phi_1 - \phi_2)} \quad (3.47)$$

$$\beta = \frac{D_2}{D} = \frac{\cot \theta_2 \sin \phi_1 - \cot \theta_1 \sin \phi_2}{\sin(\phi_1 - \phi_2)} \quad (3.48)$$

To apply the results that were achieved so far for computing the geodesic via python, one final step needs to be taken, expressing the geodesic term of the coordinate parameter ϕ and compute the corresponding arc-length parameter $s(\phi)$. This is not an arc-length parametrization; instead, the arc length is obtained afterwards via numerical integration.

Thus, ds is redefined in the next step as a function of ϕ . This can be achieved by connecting s to ϕ , stating: ds is $\frac{ds}{d\phi}$. This expresses ds such that it answers the question: how much does s change when ϕ changes —

which is exactly the derivative we need. This leads to a differential equation that cannot be solved analytically (at least not as far as I know), but can be solved via numerical integration. The following steps prepare this numerical integration.

$$\frac{ds}{d\phi} = R\sqrt{\left(\frac{d\theta}{d\phi}\right)^2 + \sin^2\theta} \quad (3.49)$$

Since it is known that $\cot\theta = A\sin(\phi - \phi')$, $\frac{d\theta}{d\phi}$ can be computed by deriving both sides by ϕ :

$$-\csc^2(\theta)\frac{d\theta}{d\phi} = A\cos(\phi - \phi') \quad (3.50)$$

$$\Rightarrow \frac{d\theta}{d\phi} = -A\cos(\phi - \phi')\sin^2\theta \quad (3.51)$$

Adding this term to formula 3.49 results in:

$$\Rightarrow R\sqrt{A^2 \cos^2(\phi - \phi') \sin^4\theta + \sin^2\theta} \quad (3.52)$$

$$= R\sqrt{(\alpha^2 \cos^2\phi + 2\alpha\cos\phi\beta\sin\phi + \beta^2 \sin^2\phi)\sin^4\theta + \sin^2\theta} \quad (3.53)$$

$$= R\sqrt{(\alpha\cos\phi + \beta\sin\phi)^2 \sin^4\theta + \sin^2\theta} \quad (3.54)$$

The function $s(\phi)$ can then be achieved via numeric integration in python using `scipy.integrate cumulative_trapezoid`, which delivers discrete steps for s between ϕ_1 and ϕ_2 .

$$\text{TrapezoidalRule} : \frac{\Delta\phi}{2}[f(\phi_0) + 2f(\phi_1) + 2f(\phi_2) + \dots + 2f(\phi_{n-1}) + f(\phi_n)] \quad (3.55)$$

There is now only one unknown left in 3.54, which is $\sin^2\theta$. To solve this very last step to computing a geodesic on a sphere using the derived formalism, the equation 3.39 $\cot\theta = \alpha\sin\phi - \beta\cos\phi$ can be used (remembering that $\cot\theta = \frac{\cos\theta}{\sin\theta}$):

$$\frac{\cos\theta}{\sin\theta} = \alpha\sin\phi - \beta\cos\phi \quad (3.56)$$

$$\cos\theta = (\alpha\sin\phi - \beta\cos\phi)\sin\theta \quad (3.57)$$

$$\cos^2\theta = (\alpha\sin\phi - \beta\cos\phi)^2 \sin^2\theta \quad (3.58)$$

$$1 = (\alpha\sin\phi - \beta\cos\phi)^2 \sin^2\theta + \sin^2\theta \quad (3.59)$$

$$1 = ((\alpha\sin\phi - \beta\cos\phi)^2 + 1)\sin^2\theta \quad (3.60)$$

$$\sin\theta = \frac{1}{\sqrt{(\alpha\sin\phi - \beta\cos\phi)^2 + 1}} \quad (3.61)$$

With this being resolved, everything needed for implementation is available.

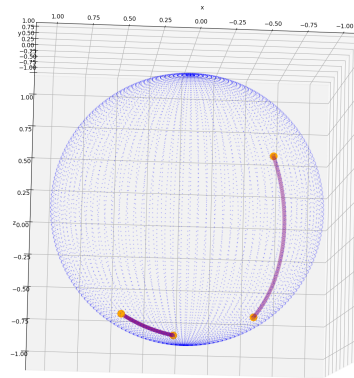


Figure 3: Two example geodesics computed in python